Francisco Gallardo Ampuero

Department of Mathematics, University of Georgia, Athens, GA 30602, USA
potitov06@uga.edu

Education

2025–present **Ph.D. in Mathematics**, University of Georgia, Athens, GA.
Advisor: Siddharth Mathur.

2023–2025 **M.S. in Mathematics**, Pontificia Universidad Católica de Chile, Santiago, Chile.
Thesis: *Brauer groups of abelian schemes*. Advisor: Siddharth Mathur.

2019–2022 **B.S. in Mathematics**, Pontificia Universidad Católica de Chile, Santiago, Chile.

Research Interests

My interests are still broad and not fully formed but mostly lie in arithmetic geometry. Lately, I've been thinking about cohomology of finite flat group schemes, abelian schemes and arithmetic hyperbolicity.

Recent talks

October 2026 *Étale cohomology comparison between affinoid adic spaces and affine schemes*, Purity for the Brauer group seminar, University of Georgia

September 2026 *Adic spaces*, Purity for the Brauer group seminar, University of Georgia

October 2025 *Unirationality of moduli space of semistable vector bundles of fixed degree and determinant on a smooth projective curve of genus* $g \geq 2$, Twisted sheaves and period-index problem seminar, University of Georgia

October 2025 *Lattices, tori and elliptic curves over the complex numbers*, Number theory seminar, University of Georgia.

Seminar Organization

Fall 2026 Co-organizer, *Purity for the Brauer group seminar*, University of Georgia (with Felipe Hernández).

Fall 2024 Co-organizer, *Hartshorne Ch. II*, Pontificia Universidad Católica de Chile (with Felipe Hernández).

Teaching

Fall 2026 Instructor of record, *MATH1113: Precalculus*, University of Georgia.

Fall 2024 TA, *Graduate algebra I*, Pontificia Universidad Católica de Chile.

Spring 2023 TA, *Abstract algebra I*, Pontificia Universidad Católica de Chile

Spring 2022 TA, *Abstract algebra II*, Pontificia Universidad Católica de Chile

Spring 2021 TA, *Abstract algebra I*, Pontificia Universidad Católica de Chile

Honors, Prizes & Scholarships

2025 *Louise Hoffmaster and Frank R. Etchberger Graduate Scholarship*, University of Georgia.

2023 *National Master's Scholarship*, Agencia Nacional de Investigación y Desarrollo (ANID), Chile. Ranked 33rd of 948 applicants.

2019 *Academic Excellence Scholarship (Matrícula de Honor)*, Pontificia Universidad Católica de Chile, for a maximum national score on the university admission test.

2018 *Bronze Medal*, XXX National Mathematical Olympiad, Sociedad de Matemática de Chile (SOMACHI).

2017 *Maximum national score*, university admission test in mathematics (PSU), DEMRE.

Conferences & Schools Attended

Nov. 2025 AGNES (Algebraic Geometry Northeast Series), University of Massachusetts Amherst.

Jan. 2025 CIMPA School *Serre's open image theorem for Galois representations attached to elliptic curves*, Universidad de Valparaíso, Chile.

Aug. 2024 CIMPA School ELGA V (Latin American School of Algebraic Geometry), IMPA, Rio de Janeiro, Brazil.

Jul. 2024 XXIV Latin American Algebra Colloquium, Pontificia Universidad Católica de Chile, Santiago.

Mar. 2024 *Moduli and Invariants in Santiago*, Pontificia Universidad Católica de Chile, Santiago.

Nov. 2022 *Geometry at the Frontier III½*, Universidad de La Frontera, Chile.

About Me

My professional life is much more than just learning mathematics on my own, though I really enjoy it. I think it is crutial to spread joy and share one's knowledge with others, mathematicians or non-mathematicians. Therefore, I think organizing talks, seminars sutdy groups, happy hours and being a good teacher/speaker is a must, not an add-on. Outside math, I enjoy walking, philosophy, chatting with people, reading, videogames, sailing, sports and good food.

Languages Spanish (native), English (fluent)